

Informe de experimento: llm_only_forced_nr

Generado el 2026-09-16 12:53 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio llm_only_forced_nr (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: llm_only_forced:nr
- Casos: case14
- Requests: generadas, N=5 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=2, parameterized=1, multistep=1, ambiguous=1
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 12:21
- Costo total: 0.1063 USD (5 items; costo medio 0.02127 USD/item)
- Items totales: 5; ok=5; con error=0; tokens totales=17105
- Solved global: 60.0% (3/5)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método llm_only_forced:nr

System prompt —860 caracteres; strategy=nr, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

Reasoning mode: you MAY write a short step-by-step reasoning in plain text before the answer. The final JSON object must be the LAST thing in the response, inside a single ```json fenced block, and it must be the only JSON object in the response. Do not use curly braces in the reasoning text.

User prompt (primer item) —6381 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

System base: 100 MVA. Every per-unit quantity below (r_pu, x_pu, b_pu) is defined on this base. Line and transformer impedances (r_pu, x_pu, b_pu) are already in per unit on this base, not ohms per km; do not convert them further. Branch thermal ratings are not defined for this case, so branch loading is undefined: report loading_percent as 0 for every branch.

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	type	zone	in_service
0	1	b	1.0	True
1	2	b	1.0	True
2	3	b	1.0	True
3	4	b	1.0	True
4	5	b	1.0	True
5	6	b	1.0	True

```

6      7      b  1.0      True
7      8      b  1.0      True
8      9      b  1.0      True
9     10      b  1.0      True
10    11      b  1.0      True
... (3 líneas omitidas de 15)

### Load Data (Load Table: net.load)

      bus      p_mw      q_mvar  in_service
0      2  21.473798  12.567615      True
1      3  85.726702  17.290948      True
2      4  45.568126  -3.717902      True
3      5   6.887322   1.449962      True
4      6  11.740497   7.861940      True
5      9  32.229269  18.135792      True
6     10   9.714580   6.260507      True
7     11   3.424253   1.761044      True
8     12   6.539657   1.715320      True
9     13  14.757610   6.340306      True
10    14  15.783969   5.296634      True

### Generator Data (Generator Table: net.gen)

      bus      p_mw  vm_pu  min_p_mw  max_p_mw  min_q_mvar  max_q_mvar  in_service
0      2  41.63323  1.045    0.0    140.0    -40.0     50.0      True
1      3   0.00000  1.010    0.0    100.0     0.0     40.0      True
2      6   0.00000  1.070    0.0    100.0    -6.0     24.0      True
3      8   0.00000  1.090    0.0    100.0    -6.0     24.0      True

### Balanced Node/External Grid (External Grid Table: net.ext_grid)

      bus  vm_pu  va_degree  in_service
0      1   1.06      0.0      True

### Line Data (Line Table: net.line)

      from_bus  to_bus      r_pu      x_pu      b_pu  parallel  in_service
0           1      2  0.01938  0.05917  0.0528      1      True
1           1      5  0.05403  0.22304  0.0492      1      True
2           2      3  0.04699  0.19797  0.0438      1      True
3           2      4  0.05811  0.17632  0.0340      1      True
4           2      5  0.05695  0.17388  0.0346      1      True
5           3      4  0.06701  0.17103  0.0128      1      True
6           4      5  0.01335  0.04211  0.0000      1      True
7           6     11  0.09498  0.19890  0.0000      1      True
8           6     12  0.12291  0.25581  0.0000      1      True
9           6     13  0.06615  0.13027  0.0000      1      True
10          9     10  0.03181  0.08450  0.0000      1      True
... (4 líneas omitidas de 16)

### Transformer Data (Transformer Table: net.trafo)

      hv_bus  lv_bus  r_pu      x_pu  shift_degree  tap_side  tap_neutral  tap_min  tap_max  tap_step_percent  in_service
0           4      7   0.0  0.20912      0.0      hv      0.0      NaN      NaN      2.2      True
1           4      9   0.0  0.55618      0.0      hv      0.0      NaN      NaN      3.1      True
2           5      6   0.0  0.25202      0.0      hv      0.0      NaN      NaN      6.8      True
3           7      8   0.0  0.17615      0.0     None      NaN      NaN      NaN      NaN      True
4           7      9   0.0  0.11001      0.0     None      NaN      NaN      NaN      NaN      True

## Task
Load case14 and disconnect the line between bus ten and bus eleven.

## Reasoning Instructions
Solve this with the Newton-Raphson power-flow procedure, worked by hand in the reasoning text before the final JSON:
1. Build the bus admittance matrix Ybus (per unit, on the stated system base) from the r_pu/x_pu/b_pu of every line and transformer (series
  ↳ admittance 1/(r_pu+j*x_pu), plus the shunt charging b_pu split half to each end bus).
2. Classify each bus as slack (ext_grid), PV (gen) or PQ (load only); PV/slack voltage magnitudes are the given setpoints, not unknowns.
3. Flat-start the unknowns: angle 0 at every non-slack bus, magnitude 1.0 p.u. at every PQ bus.
4. At each iteration: compute the power mismatch (Delta P at every non-slack bus, Delta Q at every PQ bus) from the current voltage
  ↳ estimate, form the Jacobian, solve for the angle/magnitude correction, and update the voltage estimate. Show the mismatch vector and
  ↳ the updated voltages at each iteration explicitly.
5. Repeat until the largest mismatch is below 1e-3 p.u. (report the iteration count), or state explicitly that you are not iterating
  ↳ further and why.
Do not skip straight to a final answer without showing at least one iteration's mismatch and update; a plausible-looking answer with no
  ↳ shown iteration is not an acceptable substitute.

## Output Requirements
Please calculate and return the following results (JSON format):

1) Voltage magnitude (p.u.) and phase angle (degrees) for each node

2) Active power (MW) and load factor (%) at the beginning of each branch

- For lines: `line_id` uses the row index of `net.line`
- For transformers: `line_id` uses the row index of 100000 + `net.trafo`

```

3) Total generation (MW), total load (MW), total losses (MW)

4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del **scoreboard_per_case**), en los 4 grupos de **metricas_pfagent_definicion** **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
llm_only_forced:nr	case14	5	–	1.06e-02	2.56	60.0 (3/5)	100.0 (5/5)	2.04e+000.0	–	n/a	3421	–	0.0 (0/5)	0.0 (0/5)	0.0	0.1063

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:nr	n/a	n/a	n/a	n/a

Solved por dificultad (%, items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
llm_only_forced:nr	100.0 (2/2)	100.0 (1/1)	0.0 (0/1)	0.0 (0/1)

Tipos de error de formulación (conteo de items)

(ningún item con *formulation_error_type*)

Errores de ejecución por método (conteo de items con error)

(ningún item con *error*)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método llm_only_forced:nr

Ejemplo: resuelto (solved=True)

```
request (case14-parameterized-001-s0, case14, difficulty=parameterized, seed=827308000): Load the IEEE 14-bus system and run an N-1
↳ contingency analysis ranked by min voltage and report the 3 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 3, "criteria": "min_voltage"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
 "converged": true,
 "bus_voltages": [
 {"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0},
 {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.98},
 {"bus_id": 3, "vm_pu": 1.01, "va_deg": -12.72},
 {"bus_id": 4, "vm_pu": 1.017, "va_deg": -10.33},
 {"bus_id": 5, "vm_pu": 1.019, "va_deg": -8.78},
 {"bus_id": 6, "vm_pu": 1.07, "va_deg": -14.22},
 {"bus_id": 7, "vm_pu": 1.062, "va_deg": -13.37},
 {"bus_id": 8, "vm_pu": 1.09, "va_deg": -13.36},
 {"bus_id": 9, "vm_pu": 1.056, "va_deg": -14.94},
 {"bus_id": 10, "vm_pu": 1.051, "va_deg": -15.1},
 {"bus_id": 11, "vm_pu": 1.057, "va_deg": -14.79},
 {"bus_id": 12, "vm_pu": 1.055, "va_deg": -15.08},
 {"bus_id": 13, "vm_pu": 1.05, "va_deg": -15.16},
 {"bus_id": 14, "vm_pu": 1.036, "va_deg": -16.04}
],
 "line_f ...[1408 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=4.20e-04; F_MAE=1.56; faithful=0.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3421;
↳ cost_usd=0.0213; wall=7.8 s
```

### Ejemplo: no resuelto

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
  "converged": true,
  "bus_voltages": [
    {"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0},
    {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.98},
    {"bus_id": 3, "vm_pu": 1.01, "va_deg": -12.73},
    {"bus_id": 4, "vm_pu": 1.001, "va_deg": -10.31},
    {"bus_id": 5, "vm_pu": 1.008, "va_deg": -8.77},
    {"bus_id": 6, "vm_pu": 1.07, "va_deg": -14.22},
    {"bus_id": 7, "vm_pu": 1.002, "va_deg": -13.37},
    {"bus_id": 8, "vm_pu": 1.09, "va_deg": -13.37},
    {"bus_id": 9, "vm_pu": 0.984, "va_deg": -14.94},
    {"bus_id": 10, "vm_pu": 0.979, "va_deg": -15.26},
    {"bus_id": 11, "vm_pu": 1.048, "va_deg": -14.22},
    {"bus_id": 12, "vm_pu": 1.036, "va_deg": -15.08},
    {"bus_id": 13, "vm_pu": 1.028, "va_deg": -15.27},
    {"bus_id": 14, "vm_pu": 0.963, "va_deg": -16.05}
  ],
  "line ...[1415 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.32e-02; F_MAE=2.52; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3409;
↳ cost_usd=0.0212; wall=7.4 s
```

Ejemplo: primer item

```
request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
 "converged": true,
 "bus_voltages": [
 {"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0},
 {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.01},
 {"bus_id": 3, "vm_pu": 1.01, "va_deg": -12.23},
 {"bus_id": 4, "vm_pu": 1.008, "va_deg": -10.31},

```

```

{"bus_id": 5, "vm_pu": 1.012, "va_deg": -8.77},
{"bus_id": 6, "vm_pu": 1.07, "va_deg": -14.13},
{"bus_id": 7, "vm_pu": 1.001, "va_deg": -13.62},
{"bus_id": 8, "vm_pu": 1.09, "va_deg": -13.62},
{"bus_id": 9, "vm_pu": 0.976, "va_deg": -14.72},
{"bus_id": 10, "vm_pu": 0.969, "va_deg": -15.34},
{"bus_id": 11, "vm_pu": 1.017, "va_deg": -14.40},
{"bus_id": 12, "vm_pu": 1.025, "va_deg": -14.92},
{"bus_id": 13, "vm_pu": 1.051, "va_deg": -15.18},
{"bus_id": 14, "vm_pu": 0.950, "va_deg": -16.42}
],
"line ...[1413 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.78e-02; F_MAE=6.77; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3450;
↪ cost_usd=0.0214; wall=7.4 s

```

## 5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- LLM-only: V\_MAE entre 1.06e-02 (llm\_only\_forced:nr/case14) y 1.06e-02 (llm\_only\_forced:nr/case14) p.u. sobre los items que devolvieron números.
- 0.0 % (0/5) de abstención en LLM-only (respuesta con `converged=false` o arrays vacíos).
- En LLM-only, 0 items no resueltos por `convergence_mismatch` y 2 por `voltage_error`.
- Ningún item terminó con `error`.
- Solved de `llm_only_forced:nr`: 60.0 % (3/5).
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- Trazabilidad de números (Faith.) media en LLM-only: 0.0 % (sin tools, todo número es no trazable salvo los copiados de la request).
- Costo total 0.1063 USD; el método más caro fue `llm_only_forced:nr` con 0.1063 USD (100 %).

## 6. Archivos

Rutas absolutas bajo `llm_only_forced_nr.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2]
↪ /power-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/report.json [leído:
↪ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr]
↪ ids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri]
↪ ds-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid]
↪ s-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo]
↪ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo]
↪ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/scoreboard.rescored.json
scoreboard_per_case.rescored.json:
↪ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/p]
↪ ower-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/scoreboard_per_case.rescored.json
traces (5 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sma]
↪ rtGrids-Code-v2/power-flow-agent/benchmarks/results_validation_nr/gpt-5.4/llm_only_forced_nr/traces

```